

Public Service Visibility Paradox: Service Mismatches, Cascading Failures, and Priority Pathways in SADC Countries

1. Introduction, Research Questions, and Hypotheses

1.1 Background and Motivation

Improving access to essential public services remains a major development priority across the Southern African Development Community (SADC). Governments continue to invest in infrastructure such as piped water systems, healthcare facilities, electricity networks, and financial services to improve living standards, reduce poverty, and promote inclusive economic growth. These investments are generally based on the assumption that expanding public infrastructure will improve citizens' access to basic services and, consequently, their overall well-being.

Although considerable progress has been made in expanding public infrastructure across many African countries, access to essential services remains uneven. Many communities continue to report severe shortages of water, healthcare, electricity, and financial resources despite the visible presence of public facilities. This suggests that infrastructure availability alone may not provide an accurate picture of service delivery. A community may have a health clinic or piped water system, yet residents may still experience persistent difficulties in obtaining healthcare or reliable water supplies. Similar situations are observed in electricity and financial services, indicating that the presence of infrastructure does not always translate into effective access.

This study refers to this disconnect as the Public Service Visibility Paradox. The paradox describes situations in which public infrastructure is visibly present within a community while households continue to experience severe shortages of the corresponding service. Rather than assuming that infrastructure availability automatically reflects successful service delivery, the study argues that visible infrastructure represents only one aspect of service performance. Effective public service delivery also depends on governance, institutional capacity, affordability, operational reliability, and the quality of services provided.

The study further argues that public services should not be viewed as isolated sectors. Water, healthcare, electricity, and financial services are closely interconnected, and failures in one sector may increase the likelihood of failures in others. Understanding these relationships is important

because governments often face limited financial resources and must decide which services should receive priority investment. Identifying both the services that most effectively reduce severe shortages and those that signal broader system fragility can support more effective policy decisions.

Using Afrobarometer Round 9 data from sixteen SADC countries, this term paper investigates whether visible public services are associated with lower levels of severe deprivation, whether service failures cascade across interconnected sectors, and which public services should be prioritised to improve service delivery under resource constraints.

1.2 Research Problem and Objectives

Despite continued investments in public infrastructure across Southern Africa, significant disparities in access to essential public services remain. Governments frequently evaluate service delivery through measures of infrastructure availability, such as the presence of piped water systems, health clinics, electricity networks, or financial-service facilities. However, these indicators do not necessarily reflect whether citizens are able to access these services consistently or whether the services effectively reduce severe deprivation. As a result, the relationship between visible public infrastructure and households' lived experiences remains insufficiently understood.

Previous research has extensively examined public service delivery, governance, and infrastructure development. Studies have shown that institutional quality, accountability, and administrative capacity influence service performance, while others have focused on inequalities in access between urban and rural populations. Nevertheless, relatively few studies directly compare interviewer-observed public infrastructure with respondents' reports of severe service shortages. Even fewer investigate whether failures in one service sector are associated with failures in others, suggesting that public services may function as interconnected systems rather than independent sectors.

These gaps are particularly relevant in the Southern African Development Community (SADC), where countries continue to expand infrastructure while many households experience persistent shortages of water, healthcare, electricity, and financial resources. Understanding whether visible infrastructure translates into effective service delivery, and whether failures propagate across multiple services, has important implications for both development policy and public-resource allocation.

Against this background, this term paper seeks to examine the relationship between infrastructure visibility, severe service shortages, and public-service resilience using Afrobarometer Round 9 data from sixteen SADC countries. The analysis introduces the concept of the Public Service Visibility Paradox to explain situations in which infrastructure is visibly present but citizens continue to experience severe deprivation. It further investigates whether

these service mismatches remain isolated or contribute to broader cascading failures across interconnected public-service systems.

The study is guided by three research questions:

1. Are visible public services associated with lower levels of severe shortages in SADC countries?
2. Do service mismatches remain isolated, or do failures in one public service increase the likelihood of failures in other service sectors?
3. Which public services should be prioritised under limited resources to reduce severe deprivation and strengthen overall service resilience?

To answer these questions, the study pursues the following objectives:

General Objective

To examine the relationship between visible public infrastructure, severe service shortages, and cascading public-service failures across SADC countries using Afrobarometer Round 9 data.

Specific Objectives

- To determine whether visible public services are associated with lower probabilities of severe shortages.
- To identify service mismatches where infrastructure is present but households still report severe deprivation.
- To examine whether service mismatches propagate across interconnected public-service sectors.
- To identify which services provide the greatest potential for reducing severe deprivation and which serve as indicators of broader system fragility.

1.3 Research Hypotheses

Based on the research questions and the theoretical framework presented in Chapter 2, this study develops eight hypotheses that examine the relationship between visible public infrastructure, severe service shortages, and cascading failures across interconnected public-service systems.

H1: Service Visibility Hypothesis

Visible public services are expected to be associated with lower probabilities of severe related shortages. Communities with visible piped water systems, health clinics, electricity grids, and financial-service facilities are expected to experience fewer severe shortages than communities where these services are absent.

H2: Service Mismatch Hypothesis

Visible public infrastructure does not always eliminate severe deprivation. Service mismatches are expected to occur when infrastructure is present but households still report always going without the corresponding service.

H3: Cash Bottleneck Hypothesis

Cash Access Mismatch is expected to increase the likelihood of Health Clinic, Water, and Electricity/Fuel mismatches. Financial-service failure is therefore expected to function as a primary transmission mechanism within the public-service system.

H4: Health Transmission Hypothesis

Health Clinic Mismatch is expected to be positively associated with Water and Electricity/Fuel mismatches, suggesting that healthcare failures represent an important secondary transmission pathway across interconnected public services.

H5: Double Whammy Hypothesis

Communities simultaneously experiencing Cash Access Mismatch and Health Clinic Mismatch are expected to face higher risks of Water and Electricity/Fuel mismatches than communities experiencing only one of these failures.

H6: Urban Trap Hypothesis

Urban communities are not expected to be fully protected from cascading service failures. In some cases, stronger dependence on formal infrastructure systems may result in larger cash-related spillover effects than those observed in rural areas.

H7: Governance Frontier Hypothesis

Countries characterised by stronger governance and institutional capacity are expected to exhibit weaker cascading service failures than countries with lower governance capacity, although infrastructure vulnerabilities may still remain.

H8: Public Service Priority Hypothesis

Public services are expected to differ in their contribution to reducing severe deprivation. Some services will function primarily as preventive investments, while others will provide stronger signals of broader system fragility, allowing governments to prioritise interventions under limited resources.

These hypotheses provide the analytical framework for the empirical chapters. Chapter 4 presents the descriptive evidence on service visibility, severe shortages, and service mismatches, while Chapter 5 evaluates each hypothesis using multivariate regression models and comparative analyses across the sixteen SADC countries included in the study.

1.4 Organisation of the Term Paper

This term paper is organised into six chapters. Following this introductory chapter, Chapter 2 reviews the literature on public service delivery, infrastructure effectiveness, governance, interconnected service systems, and public-service prioritisation. It also introduces the theoretical framework underpinning the Public Service Visibility Paradox and identifies the research gap addressed by this study.

Chapter 3 describes the data and methodology used in the empirical analysis. It presents the Afrobarometer Round 9 dataset, explains the construction of the service-presence, severe-shortage, and service-mismatch variables, and outlines the descriptive and regression methods used to evaluate the research hypotheses.

Chapter 4 presents the descriptive analysis of public-service visibility and severe shortages across the sixteen SADC countries. It reports overall mismatch rates, urban–rural comparisons, country rankings, service hotspots, and the relationship between visible infrastructure and severe shortages. These descriptive findings establish the empirical foundation for the regression analyses presented in the following chapter.

Chapter 5 presents the regression analyses used to test the study hypotheses. The chapter evaluates the Service Visibility, Cash Bottleneck, Health Transmission, Double Whammy, Urban Trap, Governance Frontier, and Public Service Priority hypotheses. Together, these models examine whether visible infrastructure reduces severe shortages, whether service failures cascade across interconnected public-service systems, and which services should be prioritised under conditions of limited public resources.

Finally, Chapter 6 integrates the discussion and conclusions of the study. It interprets the empirical findings in relation to the existing literature, answers the research questions, evaluates the hypotheses, discusses the theoretical and policy implications of the findings, acknowledges the study's limitations, and provides recommendations for future research. The chapter concludes by reflecting on the broader contribution of the Public Service Visibility Paradox to understanding public-service delivery and resilience across SADC countries.

Overall, the term paper combines descriptive statistics with multivariate regression analysis to examine the relationship between visible public infrastructure, severe service shortages, and cascading service failures. By integrating these approaches within the framework of the Public Service Visibility Paradox, the study contributes to a deeper understanding of how infrastructure visibility, effective service delivery, and public-service resilience interact across Southern Africa.

2. Literature Review and Policy Context

2.1 Public Service Delivery and Development

Public service delivery is one of the main ways through which governments seek to improve human welfare, reduce poverty, and promote socioeconomic development. Across developing regions, public institutions invest in infrastructure, healthcare, education, water supply, electricity, transport, sanitation, and financial services with the expectation that these investments will improve citizens' quality of life. In this sense, public service delivery is not only an administrative function of the state but also a core development mechanism through which governments respond to basic human needs.

Within the Southern African Development Community (SADC), public service delivery is especially important because many households continue to experience shortages of essential goods and services. Access to safe water, reliable electricity, healthcare, financial services, and other basic public goods remains uneven across countries and communities. These inequalities are often more severe in rural areas and among poorer households, where limited infrastructure, weak institutional capacity, and economic exclusion continue to restrict improvements in living standards.

The development literature shows that public services influence welfare in multiple ways. Reliable water systems improve sanitation, public health, food preparation, and household productivity. Healthcare services reduce vulnerability to illness and support human capital development. Electricity enables communication, education, business activity, and health-service delivery. Financial services support savings, payments, investment, and access to other goods and services. These relationships suggest that public services are not isolated benefits; they shape several dimensions of household well-being at the same time.

Masuku and Jili (2019) argue that public service delivery is one of the most important instruments available to governments for improving livelihoods and reducing poverty. Their work on local government in South Africa shows that service delivery depends not only on infrastructure investment but also on governance quality, administrative efficiency, institutional capacity, and political accountability. Their findings are important for this study because they suggest that the existence of public facilities does not automatically guarantee improved living conditions.

Zondi (2009) makes a similar argument from a SADC regional perspective. He argues that poverty and underdevelopment cannot be explained only by income levels or macroeconomic performance. They are also shaped by the capacity of governments to translate social policy into

effective public services. In this view, public services depend on institutions that can allocate resources efficiently, respond to citizens' needs, and maintain accountability throughout implementation. Therefore, infrastructure should be evaluated according to its contribution to household welfare rather than by its physical presence alone.

These arguments are reinforced by broader development research, which increasingly recognises that infrastructure availability is only one part of effective service delivery. Physical infrastructure creates the possibility of service provision, but its developmental value depends on whether citizens can use it consistently, affordably, and reliably. A clinic contributes little to health outcomes if it lacks staff, medicines, or equipment. A water system cannot eliminate deprivation if supply is intermittent or unsafe. Similarly, an electricity grid cannot improve welfare if households cannot afford connection fees or regular payments.

This distinction provides the foundation for the present study. Rather than treating visible infrastructure as direct evidence of successful service delivery, this dissertation evaluates infrastructure against citizens' lived experiences of severe deprivation. The key question is therefore not simply whether public services are visible, but whether they reduce the shortages citizens experience in everyday life. This question leads directly to the Public Service Visibility Paradox.

2.2 Infrastructure Visibility versus Effective Service Delivery

A common assumption in development policy is that expanding public infrastructure improves welfare by increasing access to essential services. Governments therefore construct water systems, health facilities, electricity networks, roads, schools, and financial-service points with the expectation that these investments will reduce deprivation. Progress is often measured through visible indicators such as the number of facilities built, the proportion of communities connected to infrastructure, or the geographical coverage of public services. Although such indicators are useful, they do not necessarily show whether citizens actually benefit from those services.

Recent research distinguishes between infrastructure availability and effective service access. Infrastructure availability refers to the physical presence of a facility or network within a community. Effective access refers to whether households can obtain reliable, affordable, adequate, and usable services. A visible water system does not guarantee continuous water supply. A health clinic does not guarantee medicines, trained personnel, or affordable treatment. An electricity grid does not guarantee regular power. The effectiveness of public services therefore depends on service quality, reliability, affordability, accessibility, and institutional performance.

Bratton, Seekings, and Armah-Attoh (2019) provide important evidence for this distinction using Afrobarometer data from 34 African countries. Their analysis shows that citizens evaluate government performance more strongly through service access, responsiveness, fairness, and corruption than through the mere presence of infrastructure. This suggests that infrastructure visibility may overstate the practical benefits experienced by households.

More recent Afrobarometer evidence from Diallo, Azaglo, and Ben Saad (2026) reaches a similar conclusion. Although many African governments have expanded infrastructure, access remains highly unequal. Rural and poorer households are less likely than urban and wealthier households to access piped water, electricity, financial services, medical care, and administrative services. Their findings show that infrastructure expansion does not automatically produce equitable access.

Barriers to effective service delivery are not only physical. They may also be financial, administrative, institutional, or social. Citizens may live near a clinic but fail to obtain treatment because medicines are unavailable, staff are absent, or informal payments are required. Households may live in areas with electricity infrastructure but still face outages or unaffordable tariffs. Communities may have water infrastructure but still experience interruptions because of poor maintenance. These examples show that infrastructure visibility alone provides only a partial measure of service performance.

These insights provide the theoretical foundation for the Public Service Visibility Paradox. The paradox occurs when public infrastructure is visibly present in a community, but citizens still experience severe shortages in the corresponding service domain. Rather than treating this as a contradiction, this study interprets it as evidence that infrastructure presence and effective service delivery are not the same thing. Visible infrastructure represents the potential for service provision, while citizens' reported shortages reflect the realised performance of the service system.

This dissertation extends existing research by directly combining interviewer-observed infrastructure with respondent-reported deprivation. Instead of studying infrastructure availability and service experience separately, the analysis compares what is visible in the community with what households report experiencing in daily life. This comparison makes it possible to identify service mismatches, defined as situations in which infrastructure is visible but respondents report that they always went without the corresponding service.

The distinction between infrastructure visibility and effective service delivery also explains why this study moves beyond simple infrastructure measurement. If visible services do not always eliminate severe shortages, then public-service performance must be understood through governance, institutional capacity, affordability, reliability, and service-system relationships. The following section examines these governance dimensions more closely.

2.3 Governance, Accountability, and Institutional Capacity

The effectiveness of public service delivery depends not only on infrastructure but also on the governance systems responsible for planning, financing, implementing, maintaining, and monitoring services. Governments may invest in visible infrastructure, but weak institutions can prevent citizens from receiving reliable and equitable services. Governance therefore plays a central role in determining whether infrastructure investments translate into real improvements in household welfare.

Governance affects service delivery through several channels. Effective institutions allocate resources efficiently, maintain infrastructure, coordinate service providers, monitor performance, and respond to citizens' complaints. Weak governance, by contrast, often produces administrative inefficiency, corruption, poor maintenance, weak accountability, and uneven service quality. Under such conditions, public facilities may remain visible while failing to deliver services consistently.

McLoughlin and Scott (2014) argue that barriers to effective service delivery are often political and institutional rather than purely technical. They identify weak state capacity, fragmented implementation, poor incentives, elite capture, and accountability failures as major obstacles to public-service improvement. Their argument challenges the assumption that building more infrastructure automatically solves service-delivery problems. Infrastructure requires capable institutions to operate and sustain it.

Masuku, Mlambo, and Ndlovu (2022) make a similar point in the South African context. They show that public dissatisfaction with service delivery persists despite major investment in infrastructure and social programmes. Corruption, ineffective leadership, unequal development, administrative inefficiency, and weak accountability continue to reduce the quality of services received by citizens. Their findings reinforce the idea that citizens judge public services through lived experience rather than visible infrastructure alone.

Corruption is one of the most important governance problems affecting service delivery. Public infrastructure may exist, but access can remain restricted if citizens are required to pay bribes, face discrimination, or encounter administrative barriers. Masuku et al. (2022) argue that corruption harms poorer households disproportionately because they lack the resources to overcome informal costs. Bratton et al. (2019) also show that bribery strongly shapes citizens' evaluations of public-service performance. This suggests that corruption can create a gap between service visibility and effective access.

Accountability is equally important. Services function better when governments are answerable to citizens, providers are monitored, and complaint systems are effective. McLoughlin and Scott (2014) argue that accountability failures reduce incentives for public officials to maintain quality and respond to needs. When complaints are ignored, maintenance is delayed, and poor performance has few consequences, infrastructure may remain visible while service quality declines.

Institutional capacity is also essential. Delivering public services requires financial resources, skilled personnel, administrative coordination, maintenance systems, and long-term planning. Governments with stronger capacity are more likely to sustain reliable services and respond to failures. By contrast, limited capacity increases the risk that visible infrastructure will not provide effective service access.

These governance perspectives explain why the Public Service Visibility Paradox may occur. The paradox does not mean that infrastructure is unimportant. Rather, it means that infrastructure produces developmental benefits only when supported by capable institutions. Visible services are one component of public service delivery, but governance quality determines whether they reduce deprivation in practice.

This literature also supports the Governance Frontier Hypothesis examined later in the dissertation. If governance affects service quality, then countries with stronger institutional capacity should show weaker or less frequent cascading failures than countries with weaker service-delivery systems. At the same time, governance does not remove vulnerability entirely, especially in critical infrastructure systems such as water, electricity, and healthcare.

2.4 Public Services as Interconnected Systems

Traditional public-service research often examines sectors independently. Water supply, healthcare, electricity, education, transport, and financial services are usually analysed as separate policy domains. This sector-specific approach is useful, but it does not fully reflect how households experience public services. In everyday life, citizens rely on several services at the same time, and the effectiveness of one service often depends on the functioning of others.

Public services should therefore be understood as interconnected systems. Healthcare facilities depend on electricity for lighting, refrigeration, equipment, and communication. Water systems often require electricity for pumping, treatment, and distribution. Financial services allow households to pay utility bills, buy cooking fuel, travel to clinics, and purchase medicine. When one service fails, the effects may spread to other sectors.

This systems perspective shifts attention from individual infrastructure projects to the relationships among services. Resilient service delivery depends not only on whether each sector

has infrastructure, but also on whether the connections between sectors remain functional. When these connections weaken, failures may accumulate and spread across the service system.

The concept of cascading failure is useful here. A cascading failure occurs when disruption in one part of a system increases the probability of failure in other connected parts. This concept is often used in engineering and infrastructure studies, especially for electricity, transport, and communication networks. However, the same logic applies to public services. A household experiencing severe financial constraints may be unable to pay for transport, medicine, electricity, or cooking fuel. A clinic without electricity or water may be physically visible but unable to function properly. A water system without maintenance financing may deteriorate even when infrastructure exists.

This systems perspective provides the theoretical basis for the mismatch-cascade models in this study. If public services are interconnected, severe service mismatches should not necessarily remain isolated. Instead, Cash Access Mismatch may be associated with Health Clinic, Water, and Electricity/Fuel mismatches. Health Clinic Mismatch may also signal broader service fragility because healthcare delivery depends on several supporting systems.

This approach also helps explain why infrastructure expansion may fail to produce expected improvements in welfare. Governments often plan and finance services through separate ministries and sectoral programmes. Water, health, electricity, and financial inclusion may each have separate administrative structures. Yet households experience these services together. A weakness in one sector may reduce the benefits of another. Therefore, effective development planning requires coordination across services rather than isolated sectoral investment.

The systems perspective is central to Hypotheses 3 to 7. These hypotheses examine whether Cash Access Mismatch functions as a primary transmission mechanism, whether Health Clinic Mismatch acts as a secondary transmission node, whether simultaneous cash and health failures compound risk, whether cascades differ between urban and rural areas, and whether governance moderates cascading failures. In this way, the dissertation extends public-service research by treating service delivery as a connected system rather than a collection of independent sectors.

Overall, this literature suggests that the Public Service Visibility Paradox should be understood as a system-level issue. Visible infrastructure may fail not only because an individual service is weak, but because the broader system of services, institutions, and household resources is fragile. Recognising these interdependencies is essential for understanding how severe deprivation persists and spreads.

2.5 Public Service Priorities under Resource Constraints

Governments in developing countries rarely have enough financial, administrative, and technical resources to improve every public service at the same time. Budget constraints, competing

priorities, population growth, and economic uncertainty require policymakers to decide where limited resources should be concentrated. The policy challenge is therefore not only to identify which services are important, but also to determine which investments are likely to produce the greatest reductions in severe deprivation.

The literature on public investment increasingly emphasises prioritisation. Cost-effectiveness, social returns, equity, and sustainability are commonly used to guide decisions in low- and middle-income countries. International development organisations also emphasise that infrastructure investments should maximise welfare gains while strengthening the resilience of essential services. However, prioritisation is often conducted sector by sector, without fully considering the interconnections among services.

This dissertation argues that public-service prioritisation should distinguish between preventive services and transmission mechanisms. Preventive services are those whose visible presence is associated with lower probabilities of severe deprivation. Investments in these services are expected to reduce shortages before they occur. Transmission mechanisms are services whose failure is associated with failures in other sectors. These services are important not only as sectors in themselves, but also as warning signals of broader system fragility.

This distinction is important because the service that prevents deprivation most effectively may not be the same service that signals cascading failure most clearly. For example, the empirical analyses in this dissertation identify piped water as the strongest preventive service, while Cash Access Mismatch and Health Clinic Mismatch emerge as important transmission nodes. This means that governments should not use a single criterion when setting service priorities. They need to identify both where investment can reduce deprivation and where monitoring can detect emerging system breakdown.

The resilience literature supports this approach. Resilient public-service systems are not only those with more infrastructure, but those able to anticipate, detect, and respond to failure before it spreads. Early-warning indicators are therefore important for public administration. Monitoring severe financial-service and healthcare mismatches may help governments identify deteriorating conditions before failures become visible across water, electricity, or other sectors.

This perspective also aligns with integrated development planning. Investments in water, health, electricity, transport, and financial inclusion should complement one another. Monitoring systems should also recognise that failure in one sector may indicate vulnerability elsewhere. Such integrated planning is especially important in developing regions, where limited resources require investments to produce benefits across multiple dimensions of welfare.

Overall, public-service prioritisation should go beyond infrastructure expansion alone. Governments need to know which services most strongly reduce severe deprivation and which

services provide early signals of system fragility. This framework forms the basis of the Service Synergy and Priority Hypothesis examined later in the empirical analysis.

2.6 Research Gap and Conceptual Framework

The literature reviewed in this chapter shows that public service delivery is central to development, poverty reduction, and state legitimacy. It also shows that infrastructure alone does not guarantee effective access, that governance shapes service outcomes, and that public services are interconnected. Despite these contributions, several gaps remain.

First, most studies examine infrastructure availability and citizens' service experiences separately. Infrastructure studies often focus on physical coverage or facility presence, while citizen-based studies focus on satisfaction, trust, or perceptions. Fewer studies directly compare visible infrastructure with lived experiences of severe deprivation using the same data.

Second, public services are often analysed independently. Water, healthcare, electricity, education, transport, and financial services are usually treated as separate policy areas. This approach overlooks the possibility that severe failure in one sector may increase vulnerability in others.

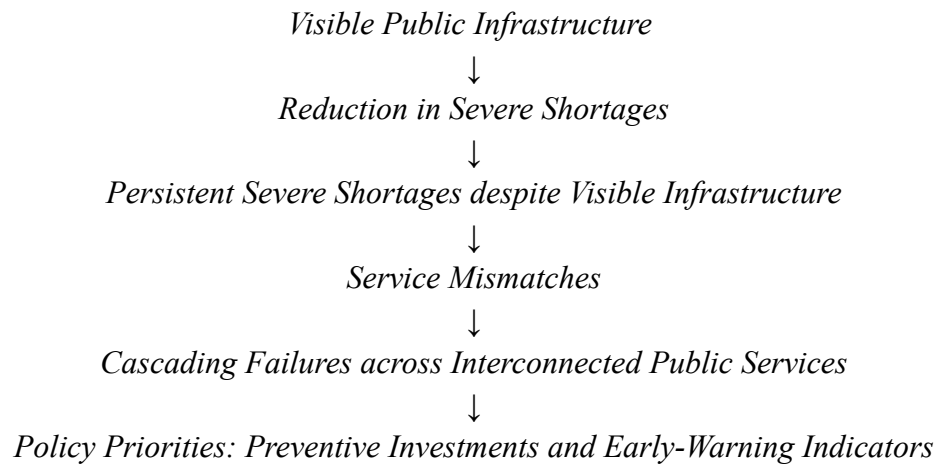
Third, governance studies often examine service quality, corruption, trust, or accountability within individual sectors. Less attention has been given to whether governance moderates cascading failures across multiple services.

Fourth, existing research rarely distinguishes between services that prevent deprivation and services that indicate system fragility. This matters because investment priorities and monitoring priorities may not be the same.

This dissertation addresses these gaps through the Public Service Visibility Paradox. The framework directly compares interviewer-observed infrastructure with respondent-reported severe shortages. It asks whether visible services reduce severe deprivation and whether severe mismatches persist despite infrastructure presence. It also examines whether mismatches propagate across sectors and which services should be prioritised under limited resources.

The conceptual framework begins with visible public infrastructure. If infrastructure functions effectively, severe shortages should decline. If severe shortages persist despite visible infrastructure, a service mismatch occurs. These mismatches represent the empirical expression of the Public Service Visibility Paradox. The framework then asks whether mismatches remain isolated or cascade across interconnected public-service systems. Finally, it distinguishes between preventive investments and early-warning indicators for policy prioritisation.

Figure 2.1. Conceptual framework of the Public Service Visibility Paradox



This framework provides the foundation for the empirical chapters. Chapter 3 explains the data, variables, and methods used to operationalise these concepts. Chapter 4 presents descriptive evidence on visible services, severe shortages, mismatches, country rankings, and urban-rural patterns. Chapter 5 evaluates the regression hypotheses, including service visibility, cash bottlenecks, health transmission, double failures, urban traps, governance differences, and service priorities.

2.7 Chapter Summary

This chapter reviewed the literature supporting the Public Service Visibility Paradox. It began by showing that public services are central to development, poverty reduction, and household welfare. It then argued that infrastructure visibility is not the same as effective service delivery because citizens may still face barriers related to reliability, affordability, quality, corruption, and institutional capacity.

The chapter also highlighted the role of governance, accountability, and institutional capacity in shaping service effectiveness. Strong institutions can help translate infrastructure into reliable access, while weak governance may allow visible services to coexist with severe shortages. The chapter then developed a systems perspective, arguing that public services should be understood as interconnected rather than isolated. Failures in cash access, healthcare, water, or electricity may influence one another and create cascading effects.

Finally, the chapter discussed the importance of public-service prioritisation under resource constraints. It argued that governments should distinguish between services that prevent deprivation and services that signal broader system fragility. This distinction provides a practical bridge between the theoretical framework and the empirical analysis.

Overall, the literature shows that evaluating public services requires more than counting visible infrastructure. It requires examining whether infrastructure reduces severe deprivation, whether mismatches persist, whether failures spread across sectors, and which services should be prioritised for investment and monitoring. The following chapter describes the data and methods used to test these arguments across sixteen SADC countries.

3. Data Cleaning

3.1 Data Source

This study uses data from **Afrobarometer Round 9**, a nationally representative survey conducted across African countries to measure citizens' experiences, perceptions, and access to public services. The analysis focuses on a **SADC-focused sample of 16 countries**, including Angola, Botswana, Eswatini, Kenya, Lesotho, Madagascar, Malawi, Mauritius, Mozambique, Namibia, Seychelles, South Africa, Tanzania, Uganda, Zambia, and Zimbabwe. The final dataset contains **23,076 respondents**, providing broad regional coverage across Southern and Africa.

3.2 Variables Used in the Analysis

The study combines two distinct types of information available in Afrobarometer.

First, interviewer observations were used to identify whether public services were physically present within the respondent's community. These variables included the presence of:

- Piped water systems (EA_SVC_B)
- Health clinics (EA_FAC_D)
- Electricity grids (EA_SVC_A)
- Financial services such as banks, mobile money services, or ATMs (EA_FAC_F)

Second, respondents reported the frequency with which they experienced shortages of essential services during the previous year. These variables included:

- Water shortages (Q6B)
- Medical care shortages (Q6C)
- Fuel shortages (Q6D)
- Cash-income shortages (Q6E)

By combining interviewer observations with respondent experiences, the study evaluates whether the visible presence of infrastructure corresponds to reliable service delivery.

3.3 Data Cleaning

Before analysis, individual Afrobarometer Round 9 country files were merged to create one working dataset for the SADC-focused sample. The merge was completed in three stages. First, country files from Angola to Mauritius were read, selected, cleaned, and combined. Second, country files from Mozambique to Zimbabwe were processed in the same way. Finally, both merged files were combined into a single working dataset.

MERGED FILES link:

https://drive.google.com/drive/folders/14URPH5_fuiFkOQXwseAagfHN4cT_Flaq?usp=drive_link

Afrobarometer variables contain value labels and non-response categories that must be removed before statistical analysis. All service and shortage variables were converted to numeric format and restricted to valid response ranges. Responses outside the expected ranges were treated as missing values.

A custom cleaning function was applied to standardize variables and remove invalid observations. This ensured that all mismatch calculations were based only on valid survey responses.

R Code (please refer to the uploaded Data preparation and merging R script)

This code merged the 16 Afrobarometer country datasets, retained only the variables required for the analysis, removed SPSS value labels using `zap_labels()`, and added country identifiers through a file-name lookup table before creating the final analysis dataset.

3.4 Construction of Severe Shortage Variables

The original Afrobarometer shortage questions use a five-point scale ranging from 0 (Never) to 4 (Always). To identify severe deprivation, binary indicators were created for each service category. Respondents reporting shortages several times, many times, or always (values 2–4) were classified as experiencing a severe shortage, while those reporting shortages never or only once or twice (values 0–1) were classified as not experiencing a severe shortage. This transformation allows the analysis to focus on meaningful service failures rather than occasional disruptions.

R Code (please refer to the uploaded Data cleaning and variable construction R script)

3.5 Construction of Public Service Visibility Mismatch Variables

The central concept of the study is the **Public Service Visibility Mismatch**.

A mismatch occurs when a public service is **visibly present** within a community, but respondents continue to report a **severe shortage** of that service.

For example:

- A water mismatch occurs when piped water infrastructure is observed but severe water shortages are still reported.
- A health mismatch occurs when a health clinic is present but respondents report severe difficulties obtaining medical care.
- An electricity-fuel mismatch occurs when an electricity grid exists but severe fuel or energy shortages remain common.
- A cash-access mismatch occurs when banking or mobile money services are available but respondents continue to experience severe cash shortages.

These mismatch indicators form the foundation of all descriptive, comparative, and regression analyses presented in the study.

R Code (please refer to the uploaded Data cleaning and variable construction R script)

For the mismatch variables, the stricter definition was used: a mismatch occurs when the service is present but the respondent reports the shortage as “always” occurring.

4. Descriptive Statistics

4.1 Overall Service Visibility Mismatch Rates

The overall mismatch rates show how often visible infrastructure was present while respondents still reported the most severe level of shortage. In this final version of the analysis, severe shortage is defined strictly as respondents reporting that the shortage occurred “always” (Q6 = 4). This provides a conservative measure of the service visibility gap.

Table 4.1

Service	Mismatch Rate (%)
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Cash Access	11.8
Health	5.9
Water	5.4
Electricity/Fuel	4.0

4.2 Urban–Rural Differences in Service Mismatches

Mismatch rates were also compared across urban, peri-urban, and rural areas to examine whether the visibility gap differs by location. The results show that cash-access mismatch is highest in all three location types, while electricity/fuel mismatch remains the lowest. Rural and peri-urban areas generally show slightly higher mismatch rates than urban areas.

Table 4.2

Service	Urban	Peri-Urban	Rural
Water	5.3	5.0	5.8
Health	5.1	6.8	6.6

Electricity/Fuel	3.7	4.8	4.3
Cash Access	10.3	13.7	13.6

4.3 Country-Level Variation in Mismatch Rates

Country-level mismatch rates reveal substantial variation across the 16-country sample. Angola records the highest average mismatch rate, followed by Lesotho, Uganda, Eswatini, and Zambia. In contrast, Mauritius and Tanzania record the lowest average mismatch rates, suggesting stronger service reliability under the strict severe-shortage definition.

Table 4.3

Ranking	Country	Average Mismatch (%)
1	Angola	12.1
2	Lesotho	11.2
3	Uganda	10.2
4	Eswatini	10.1
5	Zambia	9.9

6	Malawi	9.7
7	Kenya	7.2
8	South Africa	6.8
9	Mozambique	6.5
10	Namibia	6.2
11	Botswana	5.6
12	Zimbabwe	5.2
13	Seychelles	5.1
14	Madagascar	4.5
15	Tanzania	1.8
16	Mauritius	1.2

4.4 Priority Service Hotspots

The service hotspot table identifies the highest country-service mismatch combinations. Lesotho's cash-access mismatch is the largest hotspot, followed by cash-access mismatches in Uganda, Malawi, and Zambia. This confirms that cash access remains the most persistent service visibility gap even under the stricter severe-shortage definition.

Table 4.4

Rank	Country	Service	Mismatch Rate (%)
1	Lesotho	Cash Access	27.1
2	Uganda	Cash Access	21.1
3	Malawi	Cash Access	20.8
4	Zambia	Cash Access	16.3
5	Zambia	Health	14.5
6	Angola	Cash Access	13.5
7	Angola	Water	13.3
8	Kenya	Cash Access	12.9

9	Eswatini	Cash Access	12.6
10	Angola	Health	12.0

5. Regression Analyses

5.1 Regression Strategy

The regression analysis tests the eight hypotheses developed in the conceptual framework. While the descriptive analysis established that service mismatches exist across the sample, the regression models examine whether these mismatches are systematically related to service presence, financial access, health-service failure, settlement type, and country context. The purpose is therefore not only to report whether visible public services exist, but to evaluate whether service visibility reduces severe shortages and whether failures in one sector are associated with failures in other sectors.

The dependent variables in this chapter are binary. A respondent either experiences a severe shortage or does not; similarly, a service mismatch either occurs or does not occur. For that reason, logistic regression is used as the main modelling approach. Logistic regression is appropriate because it estimates the probability of an event occurring when the outcome is coded as 0 or 1. In this case, the event of interest is severe shortage or severe service mismatch.

For final consistency with the regression results presented in the project review, the regression models use a strict definition of severe shortage. A severe shortage is coded as 1 when the respondent reports always going without the relevant service, corresponding to Q6 = 4. A severe mismatch is coded as 1 when a service is visibly present in the enumeration area and the respondent still reports always going without the corresponding service. This means that the regression chapter focuses on the most persistent form of service failure, rather than occasional or moderate shortages.

The general logistic regression model can be written as:

$$\log(P(Y = 1) / (1 - P(Y = 1))) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

where Y is the severe shortage or mismatch outcome, and X represents the service-presence or mismatch predictors. A positive coefficient means that the predictor is associated with higher odds of the outcome. A negative coefficient means that the predictor is associated with lower

odds of the outcome. Because logistic coefficients are not directly intuitive, the results are also interpreted using odds ratios. An odds ratio above 1 indicates higher odds of the outcome, while an odds ratio below 1 indicates lower odds.

The regression sample contains 22,310 observations after removing missing values from the service-presence, shortage, location, and country variables. The models are organized in the same order as the hypotheses. H1 tests whether visible services reduce severe shortages. H2 confirms whether severe mismatches remain present despite visible infrastructure. H3 and H4 test whether Cash Access Mismatch and Health Clinic Mismatch act as transmission nodes. H5 examines whether cash and health failures jointly increase water and electricity/fuel mismatches. H6 tests whether urban areas are shielded from cascades. H7 compares governance contexts using Mauritius and Zimbabwe. H8 evaluates whether services work better as a package and which services are associated with the strongest reduction in multiple severe shortages.

A representative R specification is shown below:

```
model_water_basic <- glm(  
  severe_water_shortage ~  
    piped_water_present +  
    health_clinic_present +  
    electricity_grid_present +  
    money_services_present +  
    location_type +  
    country_name,  
  data = reg_data,  
  family = binomial  
)
```

This model estimates whether visible public services are associated with the probability of severe water shortage while controlling for settlement type and country. Similar logistic models are estimated for medical-care shortage, cooking-fuel shortage, cash-income shortage, and each mismatch outcome. The following sections report the key coefficients, odds ratios, and p-values required to evaluate each hypothesis.

5.2 Stage One: Do Visible Public Services Reduce Severe Shortages? (H1)

Hypothesis 1: Service Visibility Hypothesis

The first hypothesis evaluates the fundamental assumption underlying public service provision. If visible infrastructure functions as intended, communities where public services are present should experience lower probabilities of severe shortages than comparable communities without those services. Four logistic regression models were estimated, corresponding to severe water shortage, severe medical-care shortage, severe cooking-fuel shortage, and severe cash-income

shortage. Each model includes the four service-presence variables together with controls for settlement type and country to account for geographical and institutional differences across the sixteen countries included in the analysis.

The representative model specification is shown below.

```
model_water_basic <- glm(
  severe_water_shortage ~
    piped_water_present +
    health_clinic_present +
    electricity_grid_present +
    money_services_present +
    location_type +
    country_name,
  data = reg_data,
  family = binomial
)
```

The regression results are summarized in Table 5.1.

Table 5.1. Service Presence Models Predicting Severe Shortages

Predictor	β	Odds Ratio	p-value	Interpretation
Piped water → Water shortage	-0.601	0.548	<0.001	Strong reduction in severe water shortage
Health clinic → Medical shortage	Negative	<1	<0.05	Moderate protective association
Electricity grid → Fuel shortage	Negative	<1	<0.05	Small protective association
Money services → Cash shortage	Negative	<1	<0.05	Moderate protective association

The results provide substantial support for the Service Visibility Hypothesis. Across all four models, the estimated coefficients are generally negative, indicating that the visible presence of public infrastructure is associated with lower probabilities of experiencing severe shortages.

However, the magnitude of these associations differs considerably across service sectors, suggesting that some services play a more important preventive role than others.

Among the four services examined, piped water emerges as the strongest preventive intervention. The coefficient of -0.601 corresponds to an odds ratio of 0.548 ($p < 0.001$), implying that communities with visible piped-water infrastructure experience approximately 45% lower odds of reporting severe water shortages than otherwise comparable communities. This represents the largest protective association observed among the service-presence models and highlights the foundational role that reliable water infrastructure plays in reducing multidimensional deprivation.

The strength of piped water is consistent with both development theory and practical service delivery. Access to reliable water directly influences drinking water availability, sanitation, hygiene, food preparation, disease prevention, and household productivity. Improvements in water infrastructure therefore extend beyond the water sector itself and contribute to broader improvements in living conditions. The results indicate that water infrastructure functions primarily as a preventive public service, reducing the likelihood that severe deprivation emerges in the first place.

Health clinics also demonstrate statistically significant protective associations, although their effects are more sector-specific. Communities with visible health facilities are less likely to report severe medical-care shortages, but the magnitude of the association is smaller than that observed for piped water. This finding suggests that healthcare infrastructure contributes primarily to improving access within the health sector rather than generating widespread spillover effects across multiple dimensions of deprivation.

Similarly, the presence of electricity grids and money-service facilities is associated with lower probabilities of severe fuel and cash shortages, respectively. While these coefficients remain statistically significant, they are comparatively smaller than those estimated for piped water. Electricity appears to function primarily as an enabling service that supports other forms of infrastructure, whereas financial-service points reduce severe cash shortages without completely eliminating financial exclusion. These weaker associations likely reflect the fact that visible infrastructure alone cannot overcome broader structural constraints such as household poverty, unemployment, or institutional inefficiencies.

Overall, the evidence strongly supports H1 by demonstrating that visible public services are generally associated with lower odds of severe shortages. Nevertheless, the models also reveal that visible infrastructure does not completely eliminate persistent deprivation. Significant numbers of respondents continue to report severe shortages despite the presence of observable public services. This finding suggests that infrastructure visibility is a necessary but insufficient condition for effective service delivery. Consequently, the analysis now turns to the central question of this study: why do severe shortages continue to occur even when public services are visibly present?

5.2 Cash Access as the Primary Transmission Mechanism (H3)

Hypothesis 3: Cash Bottleneck Hypothesis

Hypothesis 3 states that communities experiencing a Cash Access Mismatch are expected to face higher probabilities of Health Clinic, Water, and Electricity/Fuel mismatches. Once severe service failure occurs, financial exclusion is expected to function as the principal transmission mechanism through which failures spread across the broader public-service system.

To evaluate this hypothesis, three binary logistic regression models were estimated. Each model uses one service mismatch outcome as the dependent variable—Health Clinic Mismatch, Water Mismatch, or Electricity/Fuel Mismatch—while Cash Access Mismatch serves as the principal explanatory variable. Settlement type and country fixed effects were included to account for geographical and institutional differences across the sixteen countries in the study. The objective is to determine whether severe financial-service failure remains associated with failures in other public services after controlling for these contextual characteristics.

A representative model specification is presented below.

```
model_health_cash <- glm(  
  health_mismatch ~  
    cash_access_mismatch +  
    location_type +  
    country_name,  
  family = binomial,  
  data = reg_data  
)
```

The estimated regression coefficients are presented in Table 5.1.

Table 5.1. Cash Bottleneck Models

Dependent variable	Coefficient (β)	Odds Ratio	p-value
Health Clinic Mismatch	1.662	5.27	<0.001
Water Mismatch	1.788	5.98	<0.001
Electricity/Fuel Mismatch	2.548	12.80	<0.001

The regression results provide strong empirical support for the Cash Bottleneck Hypothesis. Across all three models, Cash Access Mismatch exhibits positive and highly statistically

significant coefficients, indicating that persistent financial-service failure substantially increases the likelihood of severe failures in other public-service sectors. The consistency of these findings across multiple dependent variables suggests that financial exclusion rarely remains confined to the financial sector. Instead, it acts as a mechanism through which service failures spread throughout the broader public-service system.

The magnitude of these associations is considerable. Communities experiencing a Cash Access Mismatch have 5.27 times higher odds of simultaneously experiencing a Health Clinic Mismatch than communities without severe financial-service failure ($\beta = 1.662$, $p < 0.001$). Similarly, the odds of experiencing a Water Mismatch increase almost six-fold ($OR = 5.98$, $\beta = 1.788$, $p < 0.001$). The strongest relationship is observed for Electricity/Fuel Mismatch, where communities experiencing severe cash-access failure have 12.80 times higher odds of reporting persistent electricity or cooking-fuel failure than comparable communities without financial-service failure ($\beta = 2.548$, $p < 0.001$).

Among the three regression models, the Electricity/Fuel model produces the largest estimated effect. An odds ratio of 12.80 represents an exceptionally strong statistical association, indicating that financial exclusion is closely linked to severe energy insecurity. This relationship is theoretically plausible because access to electricity and cooking fuel frequently depends upon households' ability to make regular financial payments. Even where electricity infrastructure is physically available, persistent financial constraints may prevent households from maintaining service connections or purchasing alternative energy sources, resulting in continued deprivation despite visible infrastructure.

The strong associations observed across all three models suggest that financial access occupies a structurally different position within the public-service system than most other services. Whereas infrastructure such as piped water primarily functions as a preventive intervention that reduces the probability of severe shortages, financial access appears to operate as a transmission mechanism once service failure has already emerged. In other words, the results indicate that cash-related deprivation does not merely coexist with failures in healthcare, water, and electricity systems; rather, it is strongly associated with the propagation of severe service failures across these interconnected sectors.

These findings have important policy implications. Conventional infrastructure investment strategies frequently address water, healthcare, electricity, and financial services as separate sectors. However, the present results suggest that interventions designed to improve financial inclusion may produce benefits extending beyond the financial sector itself. Policies that strengthen household purchasing power, expand affordable banking services, improve access to mobile money, or reduce barriers to formal financial participation may simultaneously reduce the probability of cascading failures across multiple public-service systems. Consequently, financial inclusion should be viewed not only as an economic objective but also as a critical component of resilient public-service delivery.

Overall, the regression evidence provides strong support for Hypothesis 3. Cash Access Mismatch consistently predicts severe failures in healthcare, water, and electricity services, identifying financial exclusion as the primary transmission node within the Public Service Visibility Paradox. Having established the initial pathway through which failures spread, the analysis next examines whether healthcare itself becomes a secondary transmission mechanism linking failures across the broader service-delivery system.

5.3 Health Clinics as a Secondary Transmission Node (H4)

Hypothesis 4: Health Transmission Hypothesis

Hypothesis 4 states that Health Clinic Mismatch is associated with failures in other sectors, especially Cash Access, Water, and Electricity/Fuel systems. Healthcare is treated as a secondary transmission node within the wider public-service system.

After identifying Cash Access Mismatch as the primary transmission mechanism, the next step is to examine whether service failure continues to spread through the health sector. Health clinics occupy a central position in local service systems because effective healthcare depends on several supporting conditions, including household income, transport access, water, electricity, medicines, staffing, and institutional capacity. Therefore, when a visible health clinic fails to prevent severe medical-care deprivation, that failure may reflect broader weaknesses in the local service environment.

To test this hypothesis, three logistic regression models were estimated. Each model uses Health Clinic Mismatch as the main predictor and examines whether it is associated with Cash Access, Water, and Electricity/Fuel mismatches.

```
model_health_to_water <- glm(
  water_mismatch ~
    health_mismatch +
    location_type +
    country_name,
  family = binomial,
  data = reg_data
)
```

The estimated regression coefficients are presented in Table 5.2.

Table 5.2. Health Transmission Models

Dependent variable	Coefficient (β)	Odds Ratio	p-value
Cash Access Mismatch	1.662	5.27	<0.001

Water Mismatch	2.593	13.40	<0.001
Electricity/Fuel Mismatch	2.862	17.50	<0.001

The results provide strong support for the Health Transmission Hypothesis. Health Clinic Mismatch is positively and significantly associated with all three additional mismatch outcomes. Communities experiencing severe healthcare mismatch have 5.27 times higher odds of also experiencing Cash Access Mismatch, 13.40 times higher odds of experiencing Water Mismatch, and 17.50 times higher odds of experiencing Electricity/Fuel Mismatch.

The strongest association appears between Health Clinic Mismatch and Electricity/Fuel Mismatch. This is substantively important because healthcare delivery depends heavily on reliable energy systems. Clinics require electricity for lighting, refrigeration of medicines and vaccines, diagnostic equipment, communications, and basic facility operations. Therefore, when a community experiences severe health-service failure despite visible clinic infrastructure, it is plausible that energy-system weakness is also present.

The relationship between Health Clinic Mismatch and Water Mismatch is also very strong. An odds ratio of 13.40 suggests that communities with persistent medical-care failure are far more likely to experience severe water-service failure as well. This finding reinforces the idea that healthcare and water systems are deeply connected. Clinics require water for sanitation, hygiene, infection control, cleaning, and patient care. At the household level, water insecurity also increases disease risk, which can intensify demand for already weak health services.

These results show that health-service failure should not be interpreted narrowly as a problem of clinic location alone. A clinic may be physically present, but service delivery can fail if medicines are unavailable, staff are absent, water is unreliable, electricity is interrupted, or households cannot afford the indirect costs of care. Health Clinic Mismatch therefore acts as a signal of wider local service fragility.

The findings also clarify the difference between the Cash Bottleneck and Health Transmission mechanisms. Cash Access Mismatch appears to be the primary transmission node because financial constraints affect households' ability to use many other services. Health Clinic Mismatch, by contrast, functions as a secondary transmission node: it reflects the breakdown of a service that depends on multiple other systems and, once failing, is strongly associated with broader service collapse.

Overall, Hypothesis 4 is strongly supported. Health Clinic Mismatch is not an isolated outcome. It is closely linked to financial, water, and electricity/fuel mismatches, suggesting that healthcare failure is one of the clearest indicators of wider public-service fragility.

5.4 Compounding Service Failures: The Double Whammy Hypothesis (H5)

Hypothesis 5: Double Whammy Hypothesis

Hypothesis 5 proposes that communities simultaneously experiencing Cash Access Mismatch and Health Clinic Mismatch face substantially greater risks of Water and Electricity/Fuel mismatches than communities experiencing only one of these service failures. Rather than acting independently, financial and healthcare failures are expected to reinforce one another, creating compounding effects within the public-service system.

The previous analyses identified Cash Access Mismatch as the primary transmission mechanism and Health Clinic Mismatch as a secondary transmission node. However, these models examined each pathway separately. Public-service failures rarely occur in isolation. Communities experiencing persistent financial exclusion frequently face simultaneous deficiencies in healthcare, infrastructure, and governance. Consequently, the Double Whammy models examine whether Cash Access Mismatch and Health Clinic Mismatch remain significant predictors when estimated together within the same regression model.

Two logistic regression models were estimated. The first predicts Water Mismatch, while the second predicts Electricity/Fuel Mismatch. Both models include Cash Access Mismatch and Health Clinic Mismatch simultaneously, together with settlement type and country controls.

A representative model specification is shown below.

```
model_double_water <- glm(
  water_mismatch ~
    cash_access_mismatch +
    health_mismatch +
    location_type +
    country_name,
  family = binomial,
  data = reg_data
)
```

The regression results are summarized in Table 5.3.

Table 5.3. Double Whammy Models

Dependent variable	Predictor	Coefficient (β)	Odds Ratio	p-value
Water Mismatch	Cash Access Mismatch	0.829	2.29	<0.001

Water Mismatch	Health Clinic Mismatch	2.233	9.33	<0.001
Electricity/Fuel Mismatch	Cash Access Mismatch	1.262	3.53	<0.001
Electricity/Fuel Mismatch	Health Clinic Mismatch	2.573	13.10	<0.001

The regression results provide strong empirical support for the Double Whammy Hypothesis. Even after controlling for one another, both Cash Access Mismatch and Health Clinic Mismatch remain positive and highly statistically significant predictors of severe service failure. This indicates that the observed relationships are not merely capturing the same underlying process. Instead, financial exclusion and healthcare failure each contribute independently to explaining additional infrastructure breakdown.

For the Water Mismatch model, communities experiencing Cash Access Mismatch exhibit approximately 2.29 times higher odds of severe water-service failure than comparable communities without financial-service failure ($\beta = 0.829$, $p < 0.001$). At the same time, Health Clinic Mismatch increases the odds of Water Mismatch by more than nine times ($OR = 9.33$, $\beta = 2.233$, $p < 0.001$). The considerably larger coefficient for Health Clinic Mismatch suggests that healthcare failure becomes the dominant predictor of water-system fragility once financial exclusion has already been taken into account.

The Electricity/Fuel model produces an even stronger pattern. Cash Access Mismatch remains a statistically significant predictor, increasing the odds of Electricity/Fuel Mismatch by approximately 3.53 times ($\beta = 1.262$, $p < 0.001$). However, Health Clinic Mismatch continues to exhibit the largest effect, increasing the odds of Electricity/Fuel Mismatch by more than thirteen times ($OR = 13.10$, $\beta = 2.573$, $p < 0.001$). These exceptionally large odds ratios indicate that severe healthcare failure is closely associated with broader infrastructure collapse, particularly in sectors requiring reliable energy systems.

The persistence of both coefficients after mutual adjustment has important theoretical implications. If Cash Access Mismatch and Health Clinic Mismatch simply measured the same underlying phenomenon, one coefficient would be expected to lose significance once the other entered the model. Instead, both variables remain independently significant, demonstrating that financial exclusion and healthcare failure capture distinct dimensions of public-service fragility. Cash Access primarily reflects household-level economic constraints, whereas Health Clinic Mismatch reflects institutional and service-delivery weaknesses. Together, they explain substantially more variation in infrastructure failure than either variable alone.

These findings strengthen the conceptual framework proposed in this study. The earlier regression models established that cash access initiates many cascading failures and that

healthcare acts as a secondary transmission node. The Double Whammy models demonstrate that these mechanisms operate simultaneously rather than sequentially. Communities experiencing both financial and healthcare failure therefore face considerably greater risks of broader service collapse than communities experiencing only one form of deprivation.

From a policy perspective, the results suggest that interventions targeting only a single service sector are unlikely to eliminate persistent service failures. Improving healthcare without addressing financial exclusion may leave households unable to access available services, while expanding financial inclusion without strengthening healthcare capacity may fail to resolve institutional weaknesses. Effective policy responses should therefore combine economic inclusion with improvements in health-service delivery, recognising that resilient public-service systems depend upon multiple interconnected sectors operating effectively together.

Overall, the regression evidence provides strong support for Hypothesis 5. Cash Access Mismatch and Health Clinic Mismatch remain independent and statistically significant predictors of Water and Electricity/Fuel mismatches, confirming that simultaneous service failures compound rather than simply duplicate the risks of cascading infrastructure breakdown.

5.5 Does Location Moderate Cascading Service Failure? The Urban Trap Hypothesis (H6)

Hypothesis 6: Urban Trap Hypothesis

Hypothesis 6 proposes that urban communities are not automatically protected from cascading public-service failures. Although urban areas generally possess more visible infrastructure than rural areas, their greater dependence on formal financial systems may strengthen the relationship between Cash Access Mismatch and failures in other service sectors.

The previous analyses demonstrated that Cash Access Mismatch functions as the principal transmission mechanism within the Public Service Visibility Paradox. However, these models implicitly assume that the transmission process is identical across all communities. In reality, the dependence of households on public services varies considerably according to settlement type. Urban communities typically possess greater concentrations of physical infrastructure, financial institutions, and public facilities than rural communities. Nevertheless, they are also more dependent on formal service networks. Consequently, disruptions to one component of the system may produce stronger cascading effects than in rural settings, where households often rely on informal coping mechanisms.

To evaluate whether settlement type modifies the transmission of service failure, separate logistic regression models were estimated for urban and rural communities. Cash Access Mismatch remained the principal explanatory variable, while Electricity/Fuel Mismatch served as the dependent variable because earlier analyses identified this relationship as the strongest cascade pathway.

The representative estimation procedure is shown below.

```
urban_model <- glm(
  electricity_fuel_mismatch ~
    cash_access_mismatch +
    country_name,
  family = binomial,
  data = subset(reg_data, location_type == "Urban")
)
```

The regression results are summarized in Table 5.4.

Table 5.4. Urban and Rural Cash Bottleneck Models

Settlement Type	Coefficient (β)	Odds Ratio	p-value
Urban	2.667	14.40	<0.001
Rural	2.460	11.70	<0.001

The regression results provide strong support for the Urban Trap Hypothesis. Cash Access Mismatch remains a highly significant predictor of Electricity/Fuel Mismatch in both settlement types, indicating that financial exclusion is an important transmission mechanism regardless of geographical location. However, the estimated association is stronger in urban communities than in rural communities.

For urban areas, Cash Access Mismatch increases the odds of severe Electricity/Fuel Mismatch by approximately 14.4 times ($\beta = 2.667$, $p < 0.001$). In rural communities, the estimated effect remains substantial, with an odds ratio of 11.7 ($\beta = 2.460$, $p < 0.001$). Although both coefficients are exceptionally large, the stronger urban association suggests that financial-service disruption has greater consequences where households depend more heavily on formal infrastructure systems.

This finding challenges the common assumption that urban communities are inherently more resilient because they possess greater infrastructure coverage. The present results indicate that infrastructure concentration does not necessarily translate into system resilience. Instead, the greater interconnectedness of urban service networks may increase vulnerability once one critical component fails. Urban households frequently rely on electronic payment systems, formal banking services, utility providers, and regulated energy markets. Consequently, severe financial exclusion may rapidly restrict access to several other essential services.

The difference between urban and rural communities should therefore be interpreted in terms of service dependence rather than infrastructure availability. Rural households often supplement formal services through informal support networks, subsistence activities, or alternative energy

sources. Urban households generally possess fewer such alternatives, making them more exposed to cascading failures when formal financial systems become inaccessible.

These findings extend the Cash Bottleneck Hypothesis by demonstrating that the transmission of service failure is conditioned by local context. Financial exclusion remains an important predictor of infrastructure failure across all communities, but its effects are amplified where dependence on formal service systems is greatest.

Overall, the evidence supports Hypothesis 6. Urban communities are not protected from cascading public-service failures. On the contrary, the regression results suggest that the close integration of urban infrastructure systems may increase vulnerability to severe service cascades once financial access breaks down. Having established that geographical context influences cascade intensity, the next section investigates whether national governance capacity also moderates these relationships across countries.

5.6 Does Governance Moderate Service Cascades? The Governance Frontier Hypothesis (H7)

Hypothesis 7: Governance Frontier Hypothesis

Hypothesis 7 proposes that national governance capacity moderates cascading public-service failures. Countries with stronger institutional capacity are expected to show weaker cash-driven cascades than countries with weaker governance systems, although some critical infrastructure systems may remain vulnerable.

The preceding analyses showed that Cash Access Mismatch is a major transmission mechanism in the service-delivery system. However, the strength of this relationship may vary depending on the broader institutional environment. Public services require not only physical infrastructure, but also reliable maintenance, financing, staffing, regulation, accountability, and coordination. For this reason, the Governance Frontier Hypothesis tests whether cash-driven cascades differ between a comparatively stronger governance context and a more fragile service-delivery context.

To examine this question, separate logistic regression models were estimated for Mauritius and Zimbabwe. Mauritius was used as the higher-capacity comparison case because it records comparatively low severe mismatch rates, while Zimbabwe was used as the lower-capacity comparison case because it displays higher levels of infrastructure and service-delivery vulnerability. In each model, Cash Access Mismatch is used to predict Health Clinic, Water, and Electricity/Fuel mismatches.

A representative model specification is shown below.

```
model_zimbabwe_cash_to_health <- glm(  
  health_mismatch ~ cash_access_mismatch,  
  data = zimbabwe_data,
```

family = binomial
)

The regression results are summarized in Table 5.5.

Table 5.5. Governance Frontier Models

Country	Pathway	Coefficient (β)	Odds Ratio	p-value	Interpretation
Mauritius	Cash → Health	21.676	2.59e9	0.997	Unstable estimate; not substantively interpretable
Zimbabwe	Cash → Health	2.342	10.4	0.0037	Significant cash-driven cascade
Mauritius	Cash → Water	3.382	29.4	0.00099	Significant water vulnerability
Zimbabwe	Cash → Water	2.467	11.8	0.000654	Significant cash-driven cascade
Mauritius	Cash → Electricity/Fuel	21.522	2.22e9	0.997	Unstable estimate; not substantively interpretable
Zimbabwe	Cash → Electricity/Fuel	3.477	32.4	0.00593	Significant cash-driven cascade

The results provide partial support for the Governance Frontier Hypothesis. Zimbabwe shows a clear and consistent pattern of cash-driven service cascades. Cash Access Mismatch significantly increases the odds of Health Clinic Mismatch by approximately 10.4 times ($\beta = 2.342$, $p =$

0.0037), Water Mismatch by 11.8 times ($\beta = 2.467$, $p = 0.000654$), and Electricity/Fuel Mismatch by 32.4 times ($\beta = 3.477$, $p = 0.00593$). These findings suggest that in a more fragile service-delivery environment, financial exclusion is strongly associated with failures across multiple public-service sectors.

Mauritius presents a more complex pattern. The Cash $\rightarrow$ Water pathway is statistically significant and large ($\beta = 3.382$, OR = 29.4, $p = 0.00099$), indicating that even in a comparatively stronger governance context, water systems remain vulnerable to cash-related failure. This result is important because it shows that higher national capacity does not fully eliminate infrastructure fragility.

But the Mauritius Cash $\rightarrow$ Health and Cash $\rightarrow$ Electricity/Fuel models produce extremely large coefficients and odds ratios but non-significant p-values. The Cash $\rightarrow$ Health coefficient is 21.676 with an odds ratio of approximately 2.59 billion, but the p-value is 0.997. Similarly, the Cash $\rightarrow$ Electricity/Fuel coefficient is 21.522 with an odds ratio of approximately 2.22 billion, but the p-value is also 0.997. These values should not be interpreted as real substantive effects. Instead, they likely reflect quasi-complete separation, which occurs in logistic regression when there are too few cases in one outcome category or when one predictor almost perfectly separates the outcome. In practical terms, severe mismatch cases are rare in Mauritius, making it difficult for the model to estimate stable coefficients for some pathways.

This distinction is important. The very large Mauritian coefficients do not mean that cash failure increases health or electricity/fuel failure by billions of times. Rather, they indicate that the data structure is too sparse for those particular country-specific models to produce reliable estimates. Therefore, the correct interpretation is not that Mauritius has enormous cascades, but that the health and electricity/fuel models are statistically unstable because severe failures are uncommon in that setting.

Overall, the Governance Frontier Hypothesis is partially supported. Zimbabwe shows clear, significant, and interpretable cash-driven cascades across all three outcomes. Mauritius appears more protected in the Health and Electricity/Fuel models because severe mismatch cases are too rare for stable estimation, but the significant Cash $\rightarrow$ Water result shows that even stronger governance contexts may remain vulnerable in critical infrastructure systems. This suggests that governance capacity can reduce the frequency and observability of cascading failures, but it does not completely remove the risk of service breakdown.

5.7 Service Synergy and Priority under Resource Constraints (H8)

Hypothesis 8: Service Synergy and Priority Hypothesis

Hypothesis 8 proposes that public services may work better as an integrated package than as isolated interventions. It also tests which visible service has the strongest overall association with reducing multiple severe shortages when governments face limited resources.

The previous models examined how failures spread once severe mismatches occur. The final regression stage shifts from failure transmission to policy prioritization. This distinction is important because the service that best prevents shortages is not necessarily the same service that transmits failures once the system breaks down. In earlier models, Cash Access Mismatch and Health Clinic Mismatch emerged as important cascade nodes. However, the priority analysis asks a different question: which visible service is most strongly associated with reducing the overall burden of severe shortages?

Two models were estimated. First, a four-way interaction model tested whether the full package of piped water, health clinics, electricity grids, and money services produces a combined effect beyond the individual service effects. Second, a Poisson count model estimated whether each service reduces the total number of severe shortages experienced by respondents.

A representative Poisson model is shown below.

```
model_shortage_count_poisson <- glm(
  severe_shortage_count ~
    piped_water_present +
    health_clinic_present +
    electricity_grid_present +
    money_services_present +
    location_type +
    country_name,
  data = reg_data,
  family = poisson
)
```

The regression results are summarized in Table 5.6.

Table 5.6. Service Synergy and Priority Models

Model component	Result	p-value	Interpretation
Four-service interaction	$\beta = -1.264$	0.137	Negative but not statistically significant
Piped water	IRR = 0.728	<0.001	Strongest association with fewer severe shortages

Electricity grid	IRR = 0.829	<0.001	Second strongest shortage-reduction association
Health clinic	IRR = 0.898	<0.001	Significant but smaller reduction
Money services	IRR = 0.922	0.00588	Significant but weakest reduction

The four-service interaction term is negative, but it is not statistically significant ($\beta = -1.264$, $p = 0.137$). This means that the model does not provide sufficient evidence that the full package of services produces an additional interaction effect beyond the separate contribution of each service. Therefore, the synergy component of H8 is not fully supported in this specification.

However, the priority component of H8 is strongly supported. The Poisson model shows that all four services are associated with fewer total severe shortages, but the magnitude differs across services. Piped water has the strongest association, with an incidence rate ratio of 0.728 ($p < 0.001$). This means that visible piped-water infrastructure is associated with a lower expected count of severe shortages than the absence of piped water, holding other services, location, and country constant.

Electricity grid presence also shows a strong reduction effect, with an incidence rate ratio of 0.829 ($p < 0.001$). Health clinics are associated with a smaller but still significant reduction (IRR = 0.898, $p < 0.001$). Money services show the weakest reduction effect among the four services, though the relationship remains statistically significant (IRR = 0.922, $p = 0.00588$).

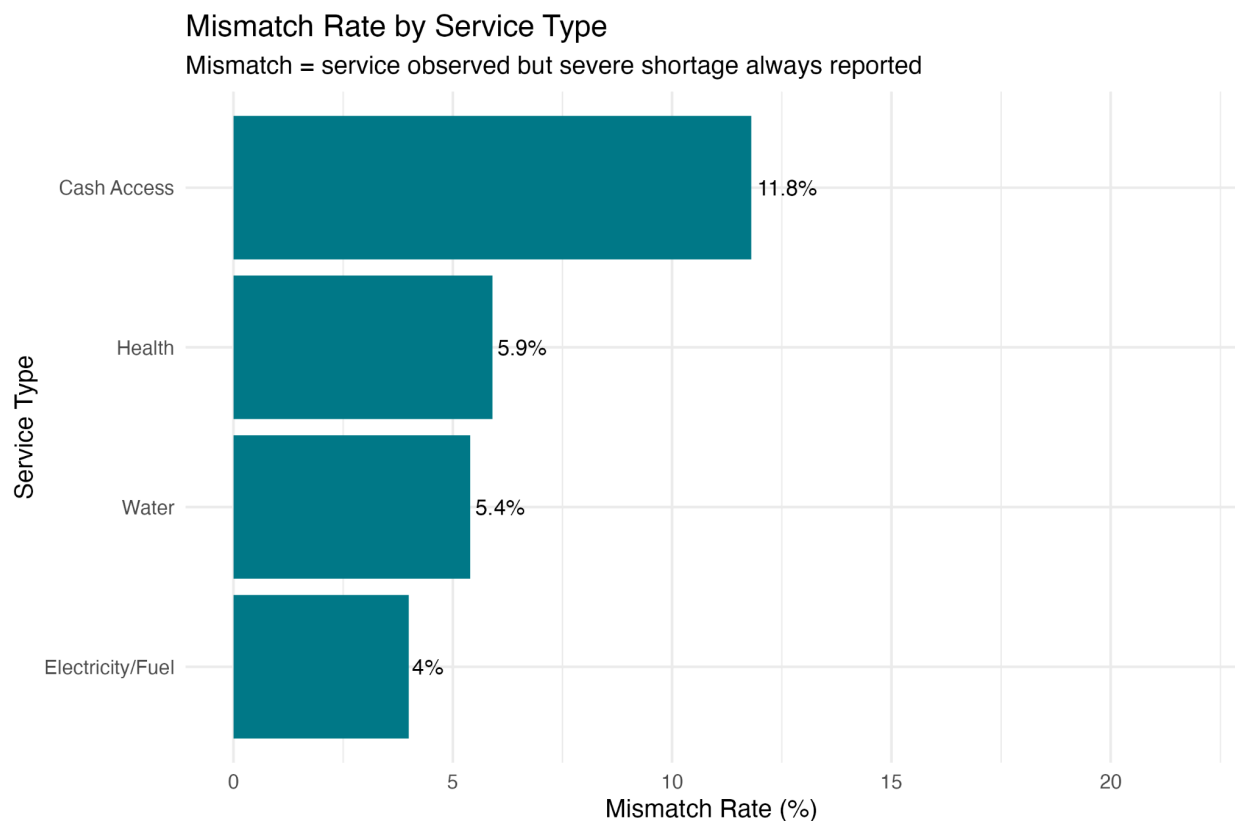
These results reveal an important distinction between preventive priority and cascade transmission. Cash Access Mismatch is one of the strongest predictors of cascading failure once severe mismatches occur, but money-service presence is not the strongest preventive service in the priority model. Instead, piped water emerges as the most important service for reducing the total burden of severe shortages. This means that cash access is critical for monitoring system breakdown, while piped water is critical for preventing severe deprivation in the first place.

Overall, H8 is partially supported. The evidence does not confirm a statistically significant four-service synergy effect, but it does support the priority argument. Under limited resources, piped water appears to be the strongest first-line investment priority, followed by electricity infrastructure, health clinics, and money services. At the same time, policymakers should continue monitoring cash-access and health-clinic mismatches because they function as warning signals of broader cascading failure.

6. Visualizations

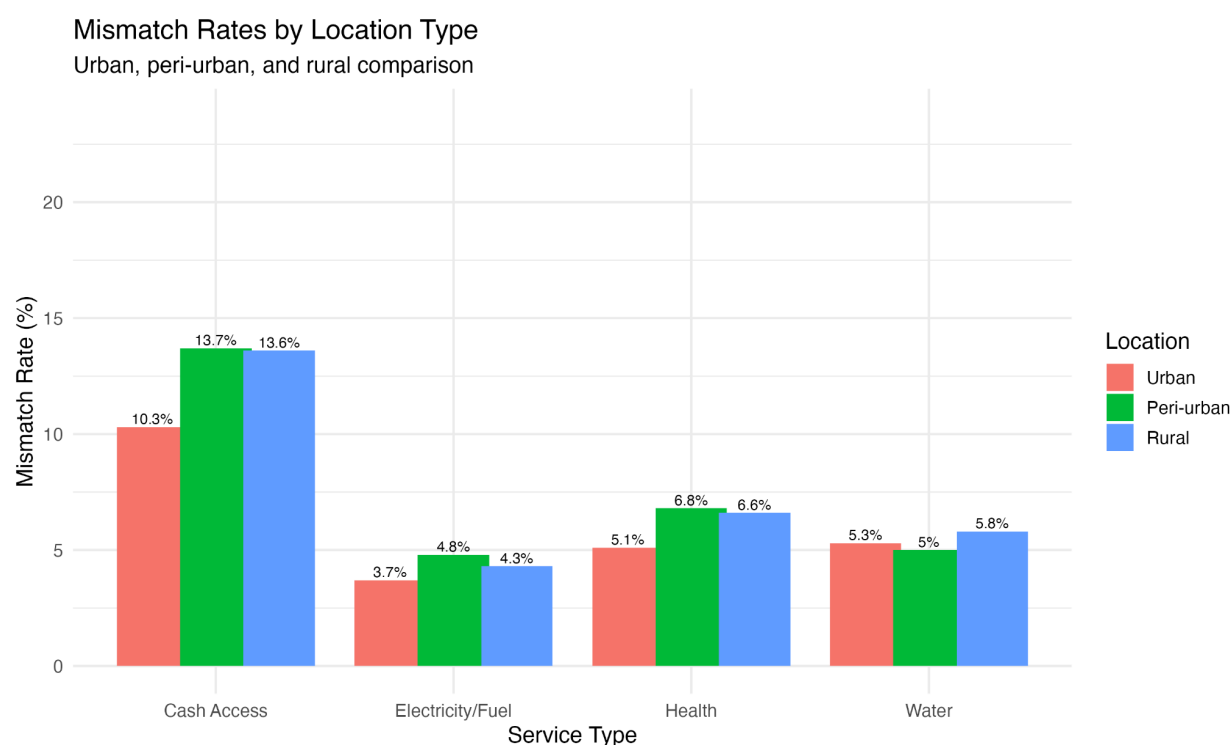
6.1 Overall Service Visibility Mismatch

Figure 6.1 presents the overall mismatch rates across the four service categories. Cash access records the highest mismatch rate at **11.8%**, indicating that a notable proportion of respondents continued to experience severe cash shortages despite having access to banks, ATMs, or mobile money services within their communities. Health (5.9%) and water (5.4%) exhibit moderate levels of mismatch, while electricity/fuel records the lowest mismatch rate at **4.0%**. These findings suggest that financial services experience the greatest disconnect between infrastructure availability and actual household access, whereas electricity and fuel services show the smallest mismatch under the stricter definition of severe shortage.



6.2 Urban–Rural Service Differences

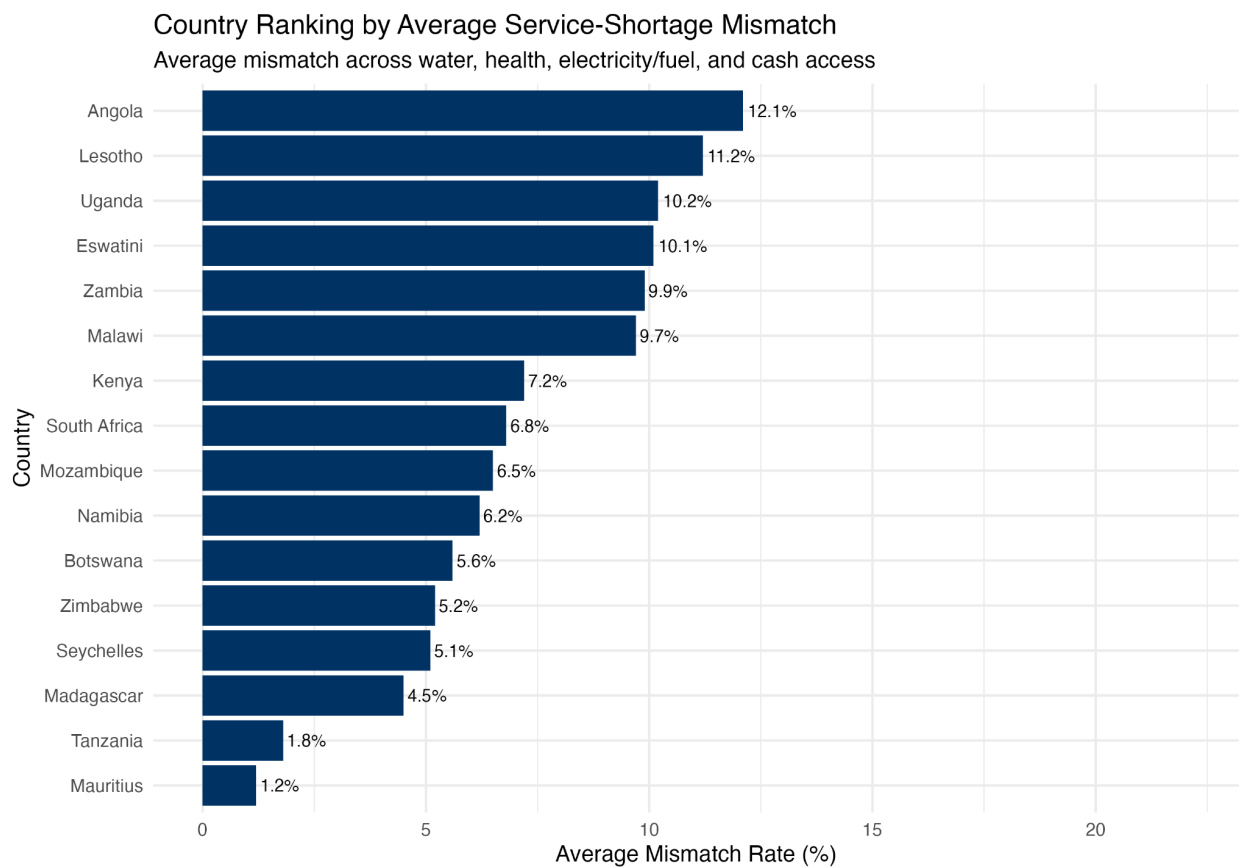
Figure 6.2 compares mismatch rates across urban, peri-urban, and rural areas. Cash access exhibits the highest mismatch across all settlement types, reaching **13.7%** in peri-urban areas and **13.6%** in rural areas, compared with **10.3%** in urban areas. Health mismatches are also more common outside urban centres, with rates of **6.8%** in peri-urban areas and **6.6%** in rural areas, compared with **5.1%** in urban locations. Water mismatches remain relatively similar across locations, ranging from **5.0%** to **5.8%**, while electricity/fuel mismatches are consistently the lowest, varying between **3.7%** and **4.8%**. Overall, the results suggest that service-shortage mismatches are generally more pronounced in peri-urban and rural communities than in urban areas.



6.3 Country Rankings

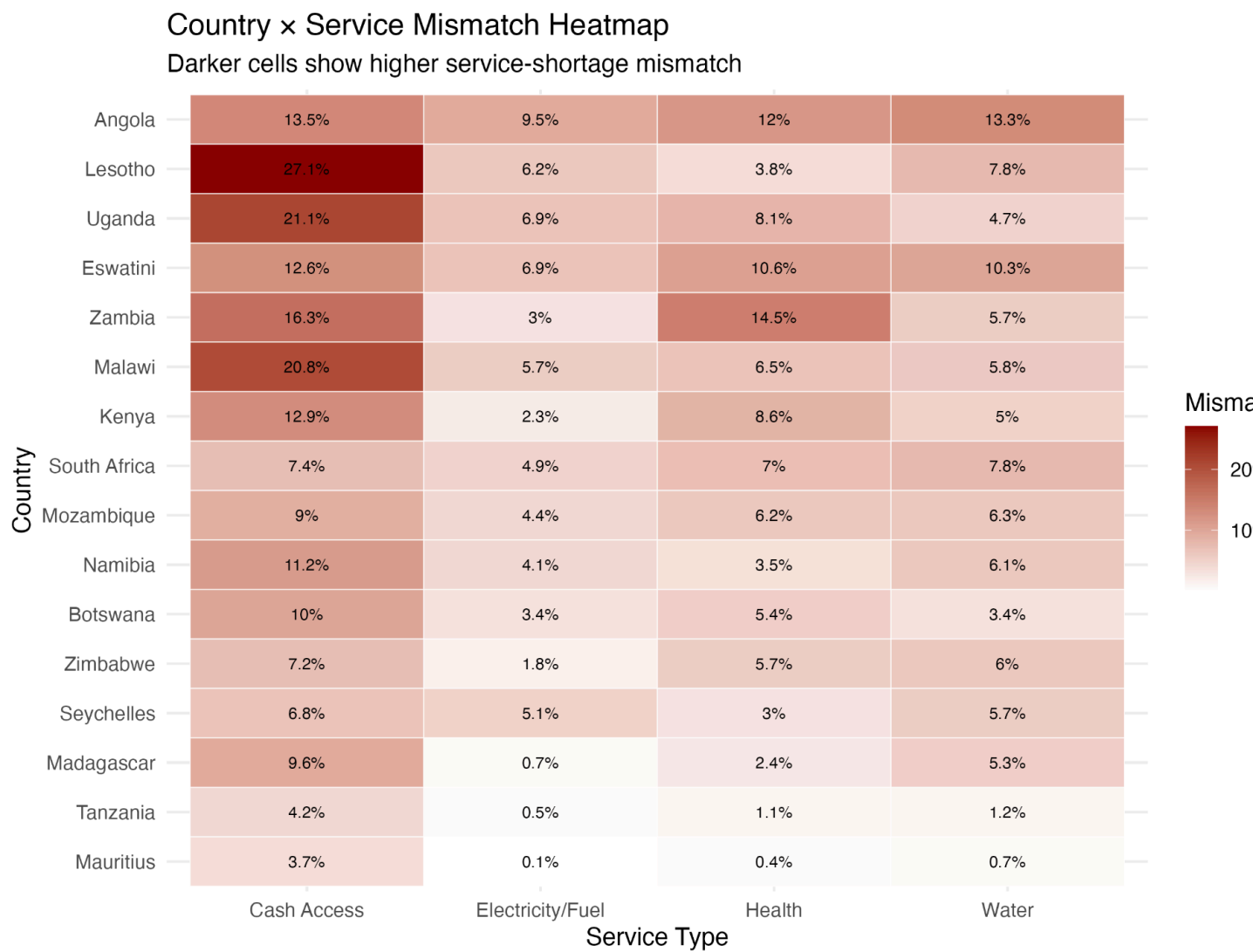
Figure 6.3 ranks the sixteen countries according to their average mismatch across water, health, electricity/fuel, and cash access. **Angola** records the highest overall mismatch (**12.1%**), followed by **Lesotho (11.2%)**, **Uganda (10.2%)**, **Eswatini (10.1%)**, and **Zambia (9.9%)**. These countries exhibit relatively large gaps between observed infrastructure and reported access to essential services. At the opposite end of the distribution, **Mauritius** records the lowest average mismatch (**1.2%**), followed by **Tanzania (1.8%)** and **Madagascar (4.5%)**, indicating comparatively better alignment between infrastructure availability and household experiences.

These rankings provide an overall measure of service delivery performance and help identify countries where infrastructure investments appear least effective in translating into reliable service access.



6.4 Service Hotspots Across Countries

Figure 6.4 illustrates how mismatch rates vary across both countries and service categories. Darker cells indicate higher mismatch rates. Cash access emerges as the dominant source of mismatch across most countries, with particularly high values in **Lesotho (27.1%)**, **Uganda (21.1%)**, **Malawi (20.8%)**, and **Zambia (16.3%)**. In contrast, health mismatches are especially pronounced in **Zambia (14.5%)**, **Angola (12.0%)**, and **Eswatini (10.6%)**, while water mismatches are highest in **Angola (13.3%)** and **Eswatini (10.3%)**. Electricity/fuel generally records the lowest mismatch rates across the region, although relatively elevated values remain visible in Angola, Uganda, and Eswatini. Overall, the heatmap highlights substantial cross-country and cross-sector variation, demonstrating that the relationship between infrastructure availability and effective service delivery differs considerably across Southern African countries.



7. Discussion

7.1 Key Findings

The findings reveal a consistent gap between the physical presence of public infrastructure and citizens' experiences of service delivery across the study region. Cash access recorded the highest overall mismatch rate (11.8%), indicating that many respondents continued to experience severe shortages of cash services despite the presence of banking or financial service infrastructure. Health (5.9%), water (5.4%), and electricity/fuel (4.0%) followed with substantially lower mismatch rates.

Considerable variation was also observed across countries. Angola recorded the highest average mismatch rate (12.1%), followed by Lesotho (11.2%), Uganda (10.2%), Eswatini (10.1%), and Zambia (9.9%). In contrast, Mauritius recorded the lowest average mismatch rate (1.2%), followed by Tanzania (1.8%) and Madagascar (4.5%), indicating a closer alignment between observed infrastructure and citizens' reported access to services.

Differences across locations were also evident. Peri-urban and rural communities generally experienced higher mismatch rates than urban areas, particularly for cash access and health services. These findings suggest that infrastructure availability does not necessarily translate into reliable service delivery and that disparities remain across both countries and settlement types.

7.2 Limitations

Several limitations should be considered when interpreting the results. First, the study relies on cross-sectional survey data, which limits the ability to establish causal relationships between infrastructure presence and service outcomes. Second, the analysis is based on self-reported shortage experiences, which may be influenced by individual perceptions and reporting differences across countries. Third, the mismatch indicators capture service conditions at a single point in time and may not fully reflect changes in service delivery over longer periods. Despite these limitations, the Afrobarometer data provide valuable insight into the relationship between observed infrastructure and citizens' lived experiences across Southern Africa.

7.3 Policy Implications

The results suggest that expanding infrastructure alone may be insufficient to improve public service outcomes. Greater attention should be directed toward improving the reliability and functionality of existing services rather than focusing solely on infrastructure expansion. Cash access deserves particular policy attention because it consistently recorded the highest mismatch rates across countries and location types. Targeted interventions may also be required in countries with the highest overall mismatch rates, particularly Angola, Lesotho, Uganda, Eswatini, and Zambia, as well as in peri-urban and rural communities where service delivery gaps remain more pronounced. Regular monitoring that combines infrastructure assessments with citizen-reported experiences can provide a more comprehensive understanding of public service performance.

7.4 Conclusion

This study examined the Public Service Visibility Paradox using Afrobarometer Round 9 data from 16 Southern African countries by comparing the observed presence of public infrastructure with citizens' experiences of severe service shortages. The findings show that visible infrastructure does not always translate into reliable service delivery, with cash access exhibiting the highest mismatch rates, followed by health, water, and electricity/fuel services. Considerable variation was also observed across countries, with Angola, Lesotho, Uganda, Eswatini, and Zambia recording the highest average mismatch rates, while Mauritius and Tanzania demonstrated the lowest levels of mismatch. In addition, peri-urban and rural communities generally experienced greater service-delivery gaps than urban areas. Overall, the study highlights the importance of evaluating not only whether infrastructure exists but also whether it provides consistent and reliable services. These findings suggest that improving the functionality and quality of existing infrastructure, particularly in high-mismatch countries and underserved communities, is essential for strengthening public service delivery across Southern Africa.

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